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## PROFILE

GIS analyst focused on spatial databases, web cartography and remote sensing. I design and build PostGIS databases with REST APIs, Leaflet-based Web GIS applications and Python pipelines for batch geodata processing. My experience runs from processing multispectral satellite imagery to network accessibility analysis across three regional capitals.

## EXPERIENCE

### GIS & Computer Vision Analyst (contractor) — CEDA Maps

June 2026 – present

*Brno, remote*

- Building a GeoAI pipeline for automatic detection and classification of road signs from mobile mapping — from data annotation through model training to production deployment
- Localising signs without precise GNSS: multi-view triangulation of observations from dozens of passes, least-squares refinement and fusion into a single entity; monocular distance estimation (Depth Anything V2, MoGe-2) as an independent source, validated by a null test and scenario synthetic data
- Clustering and deduplication of detections across multiple passes (DBSCAN, ST\_ClusterDBSCAN) and spatial analysis of routes over StreetNet data in PostGIS
- Python, PyTorch, Ultralytics YOLOv8/OBB, SAHI, ONNX Runtime, OpenCV, Open3D, laspy, COLMAP, CVAT, PostGIS, GeoPandas, Shapely

### Co-founder and technical lead — VečerkaPlus

April 2026 – present

*Frýdek-Místek, Czechia · vecerkaplus.cz*

- Co-owner and technical lead of a late-night drinks and snacks delivery service in Frýdek-Místek — the entire web platform built from scratch on React, Supabase and Vercel
- Delivery zone system built on spatial analysis: 1,093 grid points via the Google Distance Matrix API, so price and availability follow real driving distance rather than straight-line distance
- GIS market-opportunity analysis over RÚIAN, the 2021 Czech Census and OSM POIs — 265,783 inhabitants across a 630 km<sup>2</sup> zone
- Supabase backend: PostgreSQL with PL/pgSQL triggers and RLS policies, Vercel Edge Functions, automated notifications via Resend, a Telegram bot and Twilio
- Alongside development I run operations — purchasing, courier logistics and growth strategy; average revenue of CZK 452 per order at a 36.5 % margin

### Web Developer & Administrator — Palacký University Olomouc

May 2023 – present

*Olomouc, Czechia*

- Running olomouckymajales.cz and meetup.upol.cz — 199,800 visitors and 452,000 page views in total
- Requested changes delivered within 24 hours and uninterrupted operation during key events; design and functional changes deployed to production with no downtime
- WordPress, YOOftheme Builder

### GIS & Remote Sensing Analyst — Skymaps s.r.o.

February 2025 – September 2025

*Czechia*

- Processing multispectral satellite imagery into thematic maps of soil potential
- Analysis and interpretation of remote sensing data: Sentinel in SNAP, GDAL, EO Browser; Python automation from batch image processing to final map outputs
- QGIS, ArcGIS Pro, SNAP, EO Browser, Python

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## SELECTED PROJECTS

### Vehicle Detection on Orthophotos

2025 – 2026

[github.com/MetrPikeska/vehicle-detection-orthophoto](https://github.com/MetrPikeska/vehicle-detection-orthophoto)

Vehicle detection on large-area orthophotos of Olomouc, fully integrated into a GIS workflow. YOLOv8-OBB with SAHI enables inference on images larger than the GPU can hold; DBSCAN clustering of detections and Voronoi-based address assignment in GeoPandas follow. An end-to-end pipeline from CVAT annotation through training and georeferencing to spatial analysis. Presented at the international POGEO 2026 conference.

Python · YOLOv8-OBB · SAHI · GeoPandas · Shapely · CUDA

### GEOTE Klima

2025

[petrmikeska.cz/geote](https://petrmikeska.cz/geote)

A full-stack Web GIS platform for visualising and querying climate and environmental data. PostGIS with spatial indexes behind a Flask REST API, and a Leaflet front end with dynamic filtering directly on the map.

PostGIS · PostgreSQL · Flask · REST API · Leaflet · JavaScript

### Park Accessibility Toolbox

2025

[github.com/MetrPikeska/park-accessibility-toolbox](https://github.com/MetrPikeska/park-accessibility-toolbox)

An ArcGIS Python toolbox for assessing pedestrian accessibility of parks and green spaces, applied to Olomouc, Brno and Ostrava in a single workflow. Network analysis via Network Analyst, automated from data input to final maps with no manual steps. Bachelor's thesis, grade B.

ArcGIS Pro · ArcPy · Network Analyst · Python

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## SKILLS

### Languages & data-bases

Python · SQL / PL-pgSQL · JavaScript / TypeScript · R · Bash · C# · PostgreSQL / PostGIS · MySQL · Supabase

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|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Computer vision &amp; machine learning</b> | PyTorch · Ultralytics YOLOv8 / OBB · SAHI · OpenCV · torchvision · Hugging Face Transformers · Depth Anything V2 · MoGe-2 · scikit-learn · ONNX Runtime · CVAT · CUDA               |
| <b>GIS &amp; spatial data</b>                 | QGIS · ArcGIS Pro / ArcPy · GDAL/OGR · GeoServer · MapServer · SNAP · GeoPandas · Shapely · pyproj · Rasterio · H3 · Folium · OGC standards · EPSG 5514 / 3045 / 4326               |
| <b>LiDAR, 3D &amp; photogrammetry</b>         | Point cloud processing (RIEGL) · Open3D · laspy · CloudCompare · Agisoft Metashape · COLMAP (SfM) · multi-view triangulation · camera rig calibration · DTM/DSM · Blender · MeshLab |
| <b>Web &amp; backend</b>                      | React · Vite · Leaflet · FastAPI · Flask · Node.js / Express · REST API · WebSocket · HTML/CSS · WordPress · Vercel                                                                 |
| <b>Tooling &amp; operations</b>               | Git / GitHub · Linux / Ubuntu / WSL · Docker · pytest · NumPy · Pandas · SciPy · Matplotlib · Claude Code · GitHub Copilot                                                          |
| <b>Domain &amp; standards</b>                 | TP 65 (Czech road sign standard) · HD maps · mobile mapping systems · GNSS and NMEA · TomTom / StreetNet · RÚIAN · Sentinel / remote sensing · INSPIRE                              |

## EDUCATION

**MSc — Geoinformatics and Cartography (in progress)** 2025 – 2027

*Palacký University Olomouc · Faculty of Science, Dept. of Geoinformatics*

- Focus: advanced spatial data processing, web cartography, remote sensing and image processing
- Master's thesis (in progress): *Evaluating LLM-generated maps against a traditional GIS workflow*

**BSc — Geoinformatics and Cartography** 2022 – 2025

*Palacký University Olomouc · Faculty of Science, Dept. of Geoinformatics*

- Bachelor's thesis: *A user GIS toolbox for assessing the accessibility of parks and green spaces for urban residents* — grade B
- GIS, remote sensing, GNSS, geostatistics, computer and thematic cartography, spatial analysis and land-use planning

**Maturita — Programming and Application Development / Internet of Things** 2018 – 2022

*Secondary School of Information Technology · Frýdek-Místek, Czechia*

- Programming, web technologies, computer networks, databases and the fundamentals of electronics and IoT

## CERTIFICATIONS

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|---------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>October 2024</b> | <b>Remote pilot certificate, category A1/A3</b> · Civil Aviation Authority of the Czech Republic / EASA                   |
| <b>May 2023</b>     | <b>Python for Everyone</b> · Esri                                                                                         |
| <b>April 2026</b>   | <b>ISSonVIS 2026 — International Spring School on Visualization</b> · Dept. of Geoinformatics, Palacký University Olomouc |

## LANGUAGES

Czech (native) · English (B2)